Derivation of Lattice Boltzmann with BGK

Defined by DxQy where x is numebr of dimensions and y is number of vectors

Starting from the Boltzmann Equation

$$\frac{dF}{dt} \equiv \frac{\partial f}{\partial t} + \vec{\xi} \cdot \nabla_{\vec{x}} f + \frac{\vec{F}}{m} \cdot \nabla_{\vec{\xi}} f = \left(\frac{\partial f}{\partial t}\right)_{coll}, \tag{1}$$

where $f(\vec{x}, \vec{\xi}, t)$ is the particle probability distribution function, $\vec{x}$ is position, $\vec{\xi}$ is velocity and t is time.

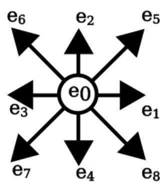
To discretize this equation we do the following

At each lattice point $\vec{x}$ we have a distribution f_i for each velocity direction $\vec{e_i}$

$$f(\vec{x}, \vec{\xi}, t) \Rightarrow f_i(\vec{x}, t) \equiv w_i f(\vec{x}, \vec{e}_i, t) \tag{2}$$

Where $\vec{e_i}$ is a velocity vector and w_i is it's weight factor

Example D2Q9:



$$\vec{e}_i = \begin{cases} (0,0) & i = 0 \\ (\pm 1, 0), (0, \pm 1) & i = 1,2,3,4 \\ (\pm 1, \pm 1) & i = 5,6,7,8 \end{cases} \tag{3}$$

Each velocity describes a direction where 0 is the center so particles which have no velocity, $[1, 2, 3, 4]$ is the cardinal directions and $[5, 6, 7, 8]$ the diagonal directions. Each direction has a weight w_i in the distribution.

We assume in one time step a packed moves exactly to the next node

$$\Delta \vec{x} = \vec{e}_i \Delta t \tag{4}$$

BGK (Bhatnagar-Gross-Krook) collision approximation where external force $\vec{F} = 0$

$$\left(\frac{\partial f}{\partial t}\right)_{coll} \approx -\frac{1}{\tau}(f - f^{eq}) \tag{5}$$

Where f^{eq} is the equilibrium density computed via a Taylor expansion. The term τ describes a unitless characteristic timescale for which the fluid relaxes locally. This timescale determines the kinematic viscosity.

Now we can write per velocity direction

$$\frac{\partial f_i}{\partial t} + \vec{e}_i \cdot \nabla_{\vec{x}} f_i = -\frac{1}{\tau}(f_i - f_i^{eq}) \tag{6}$$

Now integrate both sides by a small time step dt' since

$$\frac{\partial f_i}{\partial t} + \vec{e}_i \cdot \nabla_{\vec{x}} f_i \equiv \frac{df_i}{dt} \tag{7}$$

$$\int_0^{\Delta t} \frac{Df_i}{Dt} dt' = \int_0^{\Delta t} -\frac{1}{\tau}(f_i - f_i^{eq}) dt' \tag{8}$$

Fundamental theorem of calculus and using the assumption $\Delta\vec{x} = \vec{e}_i \Delta t$:

$$f_i(\vec{x} + \vec{e}_i \Delta t, t + \Delta t) - f_i(\vec{x}, t) = \text{Collision Integral} \tag{9}$$

Assuming that the collision term is constant over Δt

$$f_i(\vec{x} + \vec{e}_i \Delta t, t + \Delta t) - f_i(\vec{x}, t) \approx -\frac{\Delta t}{\tau}(f_i(\vec{x}, t) - f_i^{eq}(\vec{x}, t)) \tag{10}$$

For our simulation units we set $\Delta t = 1$

$$f_i(\vec{x} + \vec{e}_i, t + 1) - f_i(\vec{x}, t) \approx -\frac{1}{\tau}(f_i(\vec{x}, t) - f_i^{eq}(\vec{x}, t)) \tag{11}$$

And we get our algorithm

Step 1. Compute the distribution for the neighbour

$$f_i^{update}(\vec{x}, t) = f_i(\vec{x}, t) - \frac{1}{\tau}(f_i(\vec{x}, t) - f_i^{eq}(\vec{x}, t)) \tag{12}$$

Step 2 Update the neigbour

$$f_i(\vec{x} + \vec{e}_i, t + 1) = f_i^{update}(\vec{x}, t) \tag{13}$$

Variation in collision term approximation but by default HemeLB uses BGK

Consequence

- Lattice boltzmann is a very local algorithm only caring about local interaction.
- Additionally it lends itself to complex geometries due to the discretization process.
- Lends itself naturally to parallel compute